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Artist Talk: Lost in Space: A Composer's Journey

Abstract

As a composer trained in the acoustic Western art-music tradition, the development of new technologies has given rise to new compositional possibilities in regards to the expressive qualities of sound, the transformation of sound through time, and the movement of sound in space. The following presentation is an overview of the artist's creative practice pertaining to personal classifications of space, the application of space in his electroacoustic composition output, and compositional processes involved.

Who am I and where have I come from?

I am currently enrolled in a creative practice Doctor of Musical Arts at Queensland Conservatorium Griffith University. The submission will be a folio of original surround-sound electroacoustic works exploring the concepts of gesture and space in music composition. My initial composition training was through the Queensland Conservatorium and my influences were fairly atypical of many composition students of the mid to late-80's being primarily the works of post-WWII European Modernists (Donatoni, Ligeti, Berio, Nono etc) and working solely with acoustic media (with not much encouragement to move outside of this sphere). I was fortunate to also be exposed to the American avant garde music scene (Cage, Feldman, Crumb, Partch) and further exploration uncovered stereophonic electronic/electroacoustic works by the likes of Oliveros, Davidovsky, and Subotnik. Concurrent with my composition apprenticeship I was indulging in the enjoyment of popular music through playing bass guitar in various original and cover bands. During this time I discovered the Fairlight CMI generated works of British sampling outfit Art of Noise. This was the inspiration I required to dabble in sample-based compositions when the Conservatorium acquired a Fairlight CMI, in addition to composing the occasional work for acoustic instruments with live electronics (delay & reverb) or tape accompaniment.

After graduating I continued to compose mostly for acoustic instruments but maintained my curiosity in digital music creation through the purchasing of a hardware sequencer, synthesiser and drum machine. Much of my output was abstract 'dance' pieces for a choreographer colleague. This MIDI equipment was later superseded by the Atari ST running Notator, followed by a Power Mac 7200 running Emagic's Logic (initially MIDI only) and Finale 2.5. It was this latter system that influenced a progressive move away from composing for acoustic media towards electronic/digital media and using the technology as a performance/presentation medium. This shift generated a number of exploratory works involving lighting effects, projections, and choreography.

Composition Techniques

My personal compositional language and technique in electroacoustic music is influenced by the traditional undergraduate composition training I received where the preconceived organisation of pitch and rhythmic structures (thematic/motivic based ideas) hold a significant importance and governing influence at the micro and macro levels of the composition. Through analysing a selection of my acoustic compositions I have identified my personal usage of thematic/motivic structures as that of 'musical gesture'. Wilson Coker explains in Music and Meaning that a musical gesture "comprises a recognizable formal unit and consists of a selection and organization of sonic and rhythmic properties in sonorous motion, which signifies other purely musical objects or non-musical objects, events, and actions" (Coker 1972:18). From this statement I have defined a musical gesture for my own purpose as being: a characteristic and identifiable element in the composition embodying pitch and rhythmic organisation, dynamic and expressive design, timbral and textural qualities, and structural implications.

Fernando Lazzetta states in his article Meaning in Musical Gesture that "gesture does not mean only movement, but a movement that can express something. Therefore, it is a movement that embodies a special meaning. It is more than a change in space, or a body action, or a mechanic activity: gesture is an expressive movement that becomes actual through temporal and spatial changes" (Lazzetta 2000:260). This last sentence holds particular importance for my interpretation of a musical gesture and the implementation of this concept in my creative practice. Composing in a surround sound electroacoustic domain means no single sound, or combination of sounds, is confined to a static locational placement but is able to be moved freely around and through different areas of the performance/listening space. This

effectively generates a different expressive meaning and quality of the musical gesture through spatial movement by comparison to the musical gesture being experienced in a static or fixed locational space.

The notions of 'space' for my creative practice have evolved from the development of an analytical framework for my own works and those of other composers. I categorise space into three distinct types, however in his lecture Music/theatre as a theatre of ideas (1990) Richard Vella argues that there are only two types of space: physical space and temporal space. Vella's physical space includes all spaces that may contain a performance (real, virtual, projected etc) and in musical terms includes the movement of sound in a stereo field, the registral space of an instrument or group of instruments (high, middle or low), and harmonic space (the intervallic distance between pitches). It is the last two elements contained in this description of physical space, with the inclusion of texture/density, that I believe warrant a separation from the physical space and categorised as 'registral space'. From a compositional perspective the notion of registral space (the vertical organisation of sonic materials, the intervallic distance between pitches, and texture/density) is a separate and independent entity from the physical or locational space. In the context of an electroacoustic composition presented in a three dimension multi-speaker performance environment these two notions of space (registral and physical) could in fact be in contradiction with one another when the normal acoustic perceptions are altered. For example a low register sound could emanate from a loudspeaker positioned at the ceiling height, and a high register sound could emanate from a loudspeaker positioned on the floor. This contradictory effect could also be achieved in a binaural processed recording. This locational and registral space contradiction confers to me the separation of these events is warranted.

From these classifications, the three notions of 'space' employed in my creative practice are:

- *Temporal Space*: the structural design, linear organisation, and motion of musical events (gestures) through time
- *Registral Space*: the registral placement of sonic material, the intervallic distance between pitched material (harmonic structures), and the textural arrangement (density) of timbres and musical gestures
- *Locational Space*: the placement and movement (including distance and perspective) of musical events within the stereo or surround sound performance/listening environment

Through self analysis, and consequently the development of an analytical framework, a number of distinct compositional techniques and processes have come to the fore that typify what may be categorised as 'my style' (whatever that is!). These compositional pre-occupations can be categorised under the banners of musical gesture, temporal space, registral space, and locational space. It is worth noting that the following list of techniques constitutes my 'compositional toolbox' and not every technique or concept is employed in each composition.

Musical Gesture

For a large proportion of my output the musical gesture is the underlying component. The character and composition of the musical gesture influences much of the work's development and design in regards to pitch centres, register contour, rhythmic motion, dynamic shape, timbral and textural design, and structure. This is very much a notion of self-similarity where there is a high degree of unity between the micro and macro elements of the work. The expressive character of the musical gesture is then heightened and transformed when moved around and through the surround sound performance space. This phenomena is not dissimilar to the effect of experience that Trevor Wishart presents regarding spatial motion that "a motion is characterised not only by its path in space but also by its behaviour in time" (Wishart 1996:222). The speed of the musical gesture's movement and the type of movement that is undertaken (direct trajectories, cyclic motion, dispersion) contributes significantly to the experienced character of the material.

Temporal Space

Being the structural design, linear organisation, and motion of musical events (gestures) through time, the development and transformation the musical gesture is achieved through formulaic and intuitive processes of repetition and variation. Compositional processes and techniques include:

Rhythm

- Cellular motives
- Cyclic/isorhythmic structures
- Additive and subtractive structures
- Mechanical and fluid

Linear Pitch

- Cellular motives/pitch-set
- Cyclic phrases/isomelos
- Harmonic derivative
- Cyclic progressions

Form & Structure

- Influenced by the design and architecture of the musical gesture

Registral Space

Is the registral placement of sonic material, the intervallic distance between pitched material (harmonic structures), and the textural arrangement (density) of timbres and musical gestures. Compositional techniques and materials include:

Harmonic Structures

- Polyinterval chords
- Hierarchical pitch organistaion/Overtones

- Linear derivative

Texture

- Mechanical and fluid layers
- Circle and line (continuum and cantus firmus)

Locational Space

Composing in a stereo, quad, or 5.1 surround sound environment the placement and movement (including distance and perspective) of musical events is paramount to the compositional framework and is often designed for the specific performance/listening experience. The positioning and movement of sonic materials is influenced by the character of the musical gesture and the desired perspective of the material (foreground, middle ground, background). The following spatial motion descriptors are taken from Wishart's On Sonic Art:

- Fixed location
- Direct motion
- Cyclic motion
- Oscillatory motion
- Counterpoint of spatial motions
  - Multi-directional
  - Multi-temporal

Composition Process

My compositional process can be categorised into two components or periods of creative time. The gestation period is a significant process in the early stages of composition where the experimentation and exploration of compositional materials occurs. It is here that decisions are made concerning what type of material will constitute the piece (pitch, duration, dynamic, timbre and texture), what techniques will be employed to manipulate the chosen material, what material will be retained and discarded after the manipulation processes, and how will the remaining material be organised in time and in some instances, locational space. During the early stages of this process the compositional intention and overall concept of the work emerges, and in most cases this will be reflected in the title of the work. The time-line of these processes might be as follows:

- concept development - title of the work.
- the development of a musical gesture through improvisation and exploration.
- analysis of the gesture in terms of pitch, rhythm, expression, timbre, texture, and registral contour.
- exploration of the gesture's spatial placement and movement
- exploration of gestural characteristics at micro and macro levels.
- development of additional material through transformational processes.
- selection and de-selection of material transformation outcomes.
- the organisation of the musical structure and material transformations

The application period is the construction of the composition through the organisation and implementation of the materials and concepts developed during the gestation period. During this period the material will be further refined and decisions made regarding the use and development of the chosen material, the structure and timing of musical events, the spatial placement and movement of gestural material, and the solving of notation and/or production challenges related to the musical elements and their application within the composition. The time-line of these processes might be as follows:

- the realisation of the concept and musical materials - writing/producing the music.
- solving issues that arise concerning the notation and/or production of the musical ideas.
- maintaining a balance between processed (pre-composed) and intuitive composition techniques.
- maintaining a sense of development , timing, and dramatic tension through reflection and adjustment of composed material.
- reviewing of recorded repertoire appropriate to the concept of the work in progress.

Production Techniques and Processes

I have been fortunate during the evolution of my compositional interest in the digital music domain that technology has been running along side, if not influencing me with its developments and advancements in MIDI control, audio capturing, and signal processing. From the acquisition of my first computer, commercial software products have served my purpose for a number of musical intentions - Coda Technology's Finale for notation based works, and Emagic's (now Apple) Logic for MIDI sequencing and audio processing. This is not to say that other pieces of software, particularly shareware applications, have not been used in some part of the composition process.

As my requirements and expectations grew, so too did the computer and software capabilities. I can classify my computer-based composition journey into three phases brought about by marked increases in computer processing power, and significant advancements in the software I was using. The first phase was MIDI based only and used external MIDI sound modules. Signal processing, filter and equalisation editing were all controlled from within the host software using a custom designed sysex editor for each piece of hardware. Phase two saw the addition of audio recording, editing, and audio signal processing to Logic's capabilities. This phase also included the exciting development and implementation of VST software instruments, real-time audio processing, and the automation of controller movements (volume, panning, filter and equalisation modifications, auxiliary send levels, etc). The third (and current) phase was the discarding of external hardware sound sources in favour of a sonically self sufficient software-based composition environment. This included the emulation of 'classic'

analogue sythesisers, the increased emergence of sample-based software instruments, and more sophisticated signal processing plugins. In addition to the computer processing power required, a significant development in this phase was the ability to easily and inexpensively create surround sound DVDs.

The desktop computer has now become my composing toolbox, my timbral encyclopaedia, and my performance ensemble.

Conclusion

The transitional journey from an acoustic compositional world to an electronic compositional world is far from over for me. Every time I sit at the computer and compose for my 'performer' I am confronted with new challenges and experiences. The shift into a technology based composition process has enabled me to explore sound at a deeper level through the creation of sound worlds incorporating sampled acoustic sounds, signal processed sounds and synthetic sounds. The greatest revelation however has been the ability to mobilise sound and project those sounds into different listening spaces. The ability to move sound through space adds an entirely new dimension to the process of composing, and consequently generates an enhanced expressive quality and meaning to the musical material.

References

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